AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318
Applied Network Security

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21/April/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **60**
- This exam consists of **7** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question (1) Choose the correct answer (10 Marks):

1. What is defined as the prevention of unauthorized modification of data?
A. Availability.
B. Integrity.
C. Authentication.
D. Authorization.

2. An active attack where one entity pretends to be a different entity is called:
A. Masquerade
B. Replay
C. Denial of service
D. Traffic analysis

3. Which of the following is NOT an active attack?
A. Session hijacking
B. Traffic analysis
C. Phishing
D. Malware injection

4. Which design principle ensures that a subject is given only the minimum privileges necessary to perform its task?
A. Separation of Privileges
B. Principle of Least Privilege
C. Complete Mediation
D. Fail-safe Defaults

5. What is the fundamental idea behind Kerckhoff's Principle?
A. The algorithm must be kept secret to ensure security.
B. A cryptosystem should be secure even if everything about the system, except the key, is public knowledge.
C. Security relies on the complexity of the hardware, not the key.
D. Only authorized people should know the algorithm's design.

6. Which attack aims to reveal information by observing patterns in encrypted data or using statistical analysis?
A. Ciphertext-Only Attack
B. Known-Plaintext Attack
C. Chosen-Plaintext Attack
D. Frequency Analysis Attack

7. What is the fundamental difference between a transposition cipher and a substitution cipher?
A. Transposition ciphers are only used for data at rest, while substitution ciphers are for data in motion.
B. Transposition ciphers rearrange the plaintext letters, while substitution ciphers replace them.
C. Transposition ciphers require a numerical key, while substitution ciphers use a keyword.
D. Transposition ciphers are polyalphabetic, while substitution ciphers are monoalphabetic.

8. Which of the following correctly describes the concept of 'diffusion' as introduced by Claude Shannon for cryptographic systems?
A. It is the process of splitting the plaintext into two halves before applying a round function.
B. It dissipates the statistical structure of the plaintext by having each plaintext symbol affect many ciphertext symbols.
C. It seeks to make the relationship between the encryption key and the ciphertext as complex as possible.
D. It ensures that each plaintext block maps to a unique ciphertext block, making the encryption reversible.

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9. What is the primary purpose of the S-box used in the Substitute Bytes transformation?
A. To simply rearrange the byte order within the State.
B. To expand the key schedule for each round.
C. To introduce non-linearity and resist cryptanalytic attacks.
D. To perform a linear mixing of bits

10. Which mode of operation encrypts each block of plaintext independently of all other blocks?
A. Cipher Feedback (CFB)
B. Cipher Block Chaining (CBC)
C. Electronic Codebook (ECB)
D. Counter (CTR)

Question (2) (20 Marks):

1. What is meant by OSI Security Architecture? (4 Marks)
2. List and briefly explain three security services defined in the OSI Security Architecture (4 Marks).
3. Explain the relationship between security services and security mechanisms (2 Marks).
4. What is a chosen-plaintext attack and why is it powerful? (2 Marks)
5. Define confusion and diffusion (2 Marks).
6. Explain how each principle improves the security of a cipher (2 Marks).
7. Define a masquerading attack and what security service affected with this attack? (2 Marks).
8. What is the purpose of the DES round function F? (2 Marks)

Question (3) (4 Marks):

Given
Keyword: KEY
Plaintext: CRYPTOGRAPHY
Encrypt the plaintext using the Vigenère cipher (detail the solution)

Question (4) (6 Marks):

Use the play fair algorithm to decrypt the ciphertext **MCYVPUHZ** using the keyword **SECURITY**

Question (5) (7 Marks):

Use the frequency analysis attack on the Caesar cipher to decrypt the following ciphertext: "**WKLV PHVVDJH LV HQFUBSWHG**".
Kowing that the most common English letter is **E**, followed by **T**, **A**, **O**, **I**, **N**.

Question (6) (10 Marks):

Given
Plaintext (4 bits): $P = \mathbf{1011}$
Key (4 bits): $K = \mathbf{1100}$
S-Box

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Input	Output	Input	Output
0000	1110	1000	0011
0001	0100	1001	1010
0010	1101	1010	0110
0011	0001	1011	1100
0100	0010	1100	0101
0101	1111	1101	1001
0110	1011	1110	0000
0111	1000	1111	0111

The cipher uses 2 rounds. Each round performs:

- AddRoundKey (XOR with key)
- SubBytes (S-Box lookup)

1. Round 1
   Compute the result of AddRoundKey
   What is the final cipher after applying SubBytes
2. Round 2
   Compute the result of AddRoundKey
   What is the final cipher after applying SubBytes
3. Change one bit (last bit 1->0) of the plaintext, repeat both rounds and compute the new ciphertext.
4. Which operation provides confusion?
5. Why does changing one bit of plaintext significantly affect the ciphertext?

Question (7) (3 Marks):

A block cipher with a block size of 128 bits (e.g., AES) is used to encrypt the following plaintext blocks:

- Plaintext blocks:
  P1=AAAA
  P2=AAAA
- Key: K (same key for all modes)
- Initialization Vector (IV): BBBB

Assume:

- EK(X) represents encryption of block X using key K
- The plaintext blocks are identical

1. Write the encryption equations for ECB and CBC modes.
2. Determine whether the ciphertext blocks will be identical or different in each mode.
3. Explain why ECB is insecure for encrypting repeated data.

End of questions

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